# Linux terminal

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# Command Line

# Welcome to the Command Line!

The **Terminal** is your direct, text-based connection to the operating system.

- **Why use it?**
  - **Power & Speed:** Execute complex tasks instantly.
  - **Automation:** Script repetitive jobs.
  - **Efficiency:** Uses minimal system resources.
  - **Industry Standard:** Essential for developers and system administrators.

**Analogy:** A GUI is a restaurant menu. The CLI is speaking directly to the chef.

# The Shell & Bash

The **shell** is the program that interprets your commands.
The terminal is the window; the shell is the brain inside.

- There are many shells, each with different features:
  - `sh` (Bourne Shell): The original, classic shell.
  - `zsh` (Z Shell): A popular modern shell with extensive customization.
  - `fish` (Friendly Interactive Shell): Focuses on being user-friendly out of the box.
  - `bash` (Bourne Again SHell): The most common shell on Linux. It's the de facto standard we will learn today.

# Linux Filesystem

## Linux Filesystem: Core Directories

The filesystem is a tree starting from the **root (`/`)**.

- `/`: The **root directory**. Everything begins here.
- `/home`: Your personal files are here (e.g., `/home/student`).
- `/bin`: Essential user **binaries** (programs like `ls`).
- `/etc`: System-wide **configuration** files.
- `/var`: **Variable** data, like system logs (`/var/log`).
- `/tmp`: For **temporary** files.

# Linux Filesystem: Software & Admin

More important locations you'll encounter.

- `/opt`: **Optional** software. Used by third-party programs you install manually (e.g., Google Chrome).
- `/usr/local`: A place for software you compile or install for all users that isn't part of the standard OS distribution. You'll often find `/usr/local/bin`.
- `/root`: The home directory for the **superuser** (root user). Do not confuse this with the `/` root directory!

# Visualizing the Filesystem Tree i

Unlike Windows (which uses `C:\`, `D:\`), Linux uses a single unified tree starting at `/`.

```
/ (Root)
├── bin (Binaries/Programs)
├── etc (Configuration)
├── home (User Personal Files)
│   └── student
│       ├── Documents
│       └── Downloads
└── var (Logs and Variable Data)
```

# Visualizing the Filesystem Tree ii

**Visualizing the path:** `/home/student/Documents`

1. Start at Root /
2. Go into home
3. Go into student
4. You are in Documents

# Visualizing the Filesystem Tree iii

## User Space

### Vendor

*Sharable*
*Can be read-only*

**/usr**
Vendor-supplied resources

- ./bin — Executable commands
- ./lib — Binaries not to be executed directly at all
- ./libexec — Binaries not to be executed directly by users
- ./share — Architecture independent data

Other directories:

- **/bin** — Link to /usr/bin
- **/lib** — Link to /usr/lib
- **/sbin** — Link to /usr/bin or /usr/sbin

### Host

*Only local*
*Always writable*

**/etc**
Configuration

**/var**
Persistent, variable data

- ./cache — Cached data
- ./lib — Variable state information
- ./log — Persistent logs
- ./spool — Queued files
- ./tmp — Temp files stored for an unspecified duration

**/root**
Root user's directory

### Assets

*Sharable*
*Always writable*

**/home**
Human users' directories

- ~/.cache — Persistent user cache data
- ~/.config — Configuration and state
- ~/.local — Per-user installed software

Other directories:

- **/opt** — Application packages, managed by the administrator
- **/srv** — Server payload, managed by the administrator

### Runtime

*Flushed on boot*
*Always writable*

**/run**
Runtime data

- ./lock — Lock files
- ./log — Runtime system logs
- ./media — Mount points for removable media
- ./user — Per-user runtime directories

**/tmp**
Small temporary files

Other directories:

- **/media** — Link to /run/media
- **/mnt** — Manually mounted resources, managed by the administrator

## Kernel Space

**/boot**
The kernel and other things used for booting up

### Boot

*Only local*
*Can be read-only*

**/dev**
Physical and special devices

**/proc**
Running processes and kernel info

**/sys**
Like /dev but better structured

### Kernel API

*Virtual*
*Interface with the kernel from user space*

Sources : man file-hierarchy(7) & man udisks(8).

Modern Linux FHS

# Hidden Files & Directories

In your home directory (~), many configuration files are "hidden" by starting with a dot (`.`). They control how your programs and shell behave.

- **Examples:**
  - `~/.bashrc`: Bash shell configuration script. This is a crucial file.
  - `~/.config`: A common directory for application settings.
  - `~/.themes` or `~/.local/share/themes`: For desktop themes.
  - `~/.gitconfig`: Your Git configuration.

# Absolute vs. Relative Paths

Understanding your "address" in the filesystem tree is vital for navigation.

- **Absolute Paths:** Always start from the **root (`/`)**.
  - Example: `/home/student/Documents`.
  - Works no matter where you currently are in the system.
- **Relative Paths:** Start from your **current working directory**.
  - Example: If you are in `/home/student`, the relative path to Documents is just `Documents`.
  - `..` (Double dot) represents the parent directory (one level up).
  - `.` (Single dot) represents the current directory itself.

# Basic navigation

# Basic Navigation: pwd and cd

Two fundamental commands for moving around.

- pwd: **P**rint **W**orking **D**irectory. Shows your current location.

  ```
  $ pwd
  /home/student
  ```

- cd: **C**hange **D**irectory. Moves you to an absolute or relative path.

  ```
  $ cd /var/log       # Move to an absolute path
  $ cd Documents      # Move to a subdirectory
  ```

# Special Navigation Shortcuts with `cd`

`cd` has several useful shortcuts for faster navigation.

- Move up one level:

  ```
  $ cd ..
  ```

- Go directly to your home directory from anywhere:

  ```
  $ cd ~
  ```

  (Or just `cd` with no arguments)

- Go back to the last directory you were in:

  ```
  $ cd -
  ```

# The Magic Key: Tab Completion i

The **Tab** key is your best friend in the terminal.

- **Auto-complete:** Type the first few letters of a command or filename and press Tab. The shell will finish typing it for you.
- **Avoid Typos:** If it doesn't complete, you might have a spelling mistake.
- **List Options:** Press Tab **twice** to see a list of all matching files or commands.

# The Magic Key: Tab Completion ii

**Example:** To enter `Documents`, just type `cd Doc` and press `Tab`.

```
$ cd Doc<TAB>
# Becomes:
$ cd Documents/
```

# Listing Directory Contents: `ls`

The `ls` command **lists** the contents of a directory. It's your eyes in the terminal.

- Use **flags** to change its behavior. The most common is `-l` for a **l**ong list format.

```
$ ls -l
-rw-r--r-- 1 student student 4096 \
Sep 19 2025 my_doc.txt
drwxr-xr-x 2 student student 4096 \
Sep 17 2025 Scripts
```

- This shows permissions, owner, size, and modification date.

# Seeing Everything with `ls -a`

How do we see those hidden configuration files?

- The `-a` flag tells `ls` to show **a**ll files.

  ```
  $ ls -a
  .   ..   .bashrc   .profile   Documents   Downloads
  ```

- You can combine flags. `ls -la` is a very common command to get a **l**ong list of **a**ll files.

# Create Folders/Files

# Creating Directories: `mkdir`

Use the `mkdir` command to **m**a**k**e a new **dir**ectory.

- **Create a single directory:**

  ```
  $ mkdir my_project
  ```

- **Create a nested structure:** The `-p` (**p**arents) flag creates the entire directory path, even if the parent directories don’t exist yet.

  ```
  $ mkdir -p Documents/Work/2025/Reports
  ```

# Viewing File Contents: `cat`, `less`, `head`

You don't always need an editor (nano) just to read a file.

- **`cat`**: Dumps the **entire** file content to the screen. Good for short files.

```
$ cat /etc/hostname
```

- **`less`**: Opens a scrollable viewer. **Essential for long files!**
  - Press q to quit.
  - Use `Arrow Keys` to scroll.

```
$ less /var/log/syslog
```

- **head**: View just the first few lines of a file.

```
$ head -n 5 system.log
```

# Creating & Editing Files: `touch` & `nano`

Once you have directories, you need files to put in them.

- **`touch`:** The fastest way to create a new, empty file.

  ```
  $ touch my_notes.txt
  ```

- **nano:** A simple, friendly terminal-based text editor.

  ```
  $ nano my_notes.txt
  ```

  - Type your text directly into the window.
  - Press `Ctrl`+X to exit.
  - Press Y to confirm you want to save, then `Enter`.

# The Pro Editor: `Vim`

While nano is great for beginners, `Vim` (Vi IMproved) is a powerful industry standard.

- It is available on almost every Linux server and is designed for speed without leaving the home row of your keyboard.
- **Modal Editing:** Vim has different “modes”:
- **Normal Mode:** For navigation and commands (default).
- **Insert Mode:** For typing text (Press `i`).
- **Command Mode:** For saving/quitting (Press `:`).
- **The Exit Strategy:** To save and quit, press `Esc`, type `:wq`, and press `Enter`. To quit without saving, use `:q!`.

# File Operations: `cp`, `mv`, and `rm`

Once you can create files, you need to know how to manage their lifecycle.

- **`cp` (Copy):** Creates a duplicate of a file or directory.
  - `$ cp file.txt backup.txt`
  - Use `-r` to copy directories recursively.
- **`mv` (Move/Rename):** Moves a file to a new location or renames it.
  - `$ mv old_name.txt new_name.txt` (Rename)
  - `$ mv file.txt Documents/` (Move)
- **`rm` (Remove):** Deletes files or directories.
  - **Warning:** There is no “Trash” bin in the terminal; deletion is permanent.
  - Use `rm -r` to delete a folder and all its contents.

# Getting User & System Information

# Getting System Information

The terminal is excellent for quickly checking system status.

- `whoami`: Shows your current username.
- `date`: Shows the current date and time.
- `uname -a`: Shows kernel and system info.
- `top`: Shows running processes in real-time (like Task Manager). Press q to quit.

# Real-time Processes: htop

While `top` is the default, `htop` provides a much more user-friendly, color-coded, and interactive interface.

- **Visual Bars:** See CPU usage per core, memory usage, and swap at a glance.
- **Interactivity:** Scroll vertically and horizontally; kill processes ($F9$) without typing PIDs.
- **Search/Filter:** Easily find specific processes ($F3$ or $F4$).

```
$ sudo apt install htop  # If not installed
$ htop
```

## Hardware Deep-Dive: `dmidecode`

`dmidecode` dumps a computer’s DMI (SMBIOS) table contents into a human-readable format.

- **Hardware Info:** Provides details about BIOS, serial numbers, RAM speeds, and motherboard slots.
- **Privilege:** Requires `sudo` because it reads system memory.

**Common Usage:**

```
# Get specific info about the system (e.g., memory)
$ sudo dmidecode -t memory
# Get the system serial number
$ sudo dmidecode -s system-serial-number
```

## The CPU through the Filesystem: `/proc/cpuinfo`

In Linux, “everything is a file.” The `/proc` directory is a virtual filesystem that acts as a window into the kernel.

- **`/proc/cpuinfo`:** contains the detailed parameters of your processor.

```
# View your CPU model, cores, and cache size
$ cat /proc/cpuinfo | grep "model name"
$ cat /proc/cpuinfo | grep "cpu MHz"
```

# Memory and Swap: `free`

To get a quick snapshot of how your RAM and Swap space are being utilized, use the `free` command.

- **-h Flag:** Displays values in **human-readable** format (GB, MB) instead of bytes.
- **Swap:** This is “virtual memory” on your disk used when physical RAM is full.

```
$ free -h
      total  used   free shared buff/cache availabl
Mem:   30Gi  10Gi  2.6Gi  606Mi       18Gi      19G
Swap: 4.0Gi 768Ki  4.0Gi
```

- **`buff/cache`:** Memory used by the kernel for optimization; it is released if applications need it.

# Users: Standard vs. Superuser

Linux is a multi-user system.

- **Standard User** (`student`): Your day-to-day account with limited privileges.
- **Superuser** (`root`): The administrator. Has complete power over the system.

To run one command with root privileges, use sudo (**S**uper**u**ser **do**).

```
# This needs admin rights, so we use sudo
$ sudo apt update
```

# Managing Users

As an administrator, you can manage user accounts from the command line.

- `sudo useradd new_user`: Creates a new user.
- `sudo passwd new_user`: Sets the password for the new user.
- `sudo userdel new_user`: Deletes a user.

# File Permissions

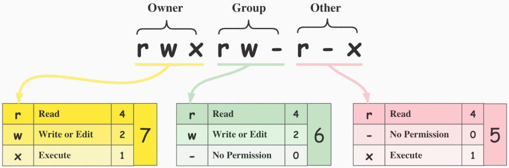
# Understanding File Permissions i

The `ls -l` command shows permissions as a 10-character string like `-rwxr-xr--`.

- **It's read in groups:** Type | Owner | Group | Others
- `r`: Permission to **read** the file.
- `w`: Permission to **write** (modify) the file.
- `x`: Permission to **execute** the file (run as a program).

# Understanding File Permissions ii

| Binary | Octal | String Representation | Permissions |
|---|---|---|---|
| 000 | 0 (0+0+0) | --- | No Permission |
| 001 | 1 (0+0+1) | --x | Execute |
| 010 | 2 (0+2+0) | -w- | Write |
| 011 | 3 (0+2+1) | -wx | Write + Execute |
| 100 | 4 (4+0+0) | r-- | Read |
| 101 | 5 (4+0+1) | r-x | Read + Execute |
| 110 | 6 (4+2+0) | rw- | Read + Write |
| 111 | 7 (4+2+1) | rwx | Read + Write + Execute |




File Permissions

# Managing Permissions with chmod

Use the chmod (**ch**ange **mod**e) command to change permissions.

- You can add (+) or remove (-) permissions for the **u**ser, **g**roup, or **o**thers.

**Example:** Make a script executable for yourself.

```
# Give the user (u) the execute (x) permission
$ chmod u+x my_script.sh
```

# Package Manager

# What is a Package Manager?

A package manager is a tool that automates the process of installing, updating, and removing software.

- It handles **dependencies** automatically, so you don’t have to install required libraries manually.
- It keeps a database of installed software, making it easy to manage.
- For Debian and Ubuntu-based systems, the primary package manager is **APT** (Advanced Package Tool).

**Analogy:** Think of `apt` as an App Store for your terminal.

## Updating Package Lists (`apt update`)

Before you install or search for anything, you should synchronize your local package list with the central software repositories.

- This command **does not** upgrade your software. It just downloads the latest list of what's available.
- This is a privileged operation, so it requires sudo.

```
# Downloads the latest package information
$ sudo apt update
```

# Searching for Packages (`apt search`)

If you're not sure of the exact name of a program, you can search for it.

- This command searches the names and descriptions of all available packages.
- You don't need `sudo` to search.

**Example:** Search for a program that shows system processes, like `htop`.

```
$ apt search htop
```

# Installing Packages (`apt install`)

Once you know the package name, you can install it.

- `apt` will automatically download and install the program and any dependencies it needs to run.
- This requires `sudo`.

**Example:** Install the `htop` utility, an interactive process viewer.

```
$ sudo apt install htop
```

After installation, you can run the program by simply typing `htop`.

# Removing Packages (`apt remove` / `apt purge`)

Removing software is just as easy as installing it. You have two main options:

1. **`apt remove`**: Uninstalls the program but leaves its configuration files behind (useful if you plan to reinstall it later).
2. **`apt purge`**: Uninstalls the program **and** deletes all of its configuration files.

**Examples:**

```
# Remove htop but keep its config files
$ sudo apt remove htop

# Remove htop and all of its config files
$ sudo apt purge htop
```

# Cron & Crontab

# Introduction to `cron` & `crontab`

**`cron`** is a system daemon (a background process) that runs scheduled tasks. These scheduled tasks are known as **"cron jobs."**

- It's the standard tool for automating repetitive tasks on a schedule.
- You manage your personal list of cron jobs using the **`crontab`** command.

**Common Uses:**

- Running a backup script every night.
- Performing system maintenance, like a weekly **ZFS scrub** or a daily **SSD trim**.
- Cleaning up temporary files.

# Understanding crontab Syntax i

A cron job consists of two parts: the **schedule** and the **command**. The schedule is defined by five fields, often represented by asterisks (*).

```
┌───────────── minute (0 - 59)
│ ┌───────────── hour (0 - 23)
│ │ ┌───────────── day of month (1 - 31)
│ │ │ ┌───────────── month (1 - 12)
│ │ │ │ ┌───────────── day of week (0 - 6) (Sunday
│ │ │ │ │
│ │ │ │ │
* * * * * /path/to/command
```

# Understanding `crontab` Syntax ii

An asterisk * means “every.” For example, an asterisk in the “hour” field means “every hour.”

For an easy way to generate the correct time string, check out: crontab.guru

# Managing Your `crontab`

You can edit, view, and remove your cron jobs with the `crontab` command and a flag.

- `crontab -e`: **Edit** your crontab file. The first time you run this, it will ask you to choose a text editor (like nano).
- `crontab -l`: **List** your currently scheduled cron jobs.
- `crontab -r`: **Remove** your entire crontab file (use with caution!).

# `crontab` Examples i

Here are some practical examples you might add using `crontab -e`.

**Example 1: Run a backup script every day at 3:30 AM.**

```
# Minute Hour Day(M) Month Day(W) Command
30 3 * * * /home/student/scripts/backup.sh
```

**Example 2: Run a system maintenance command every Sunday at 4:00 AM.** This example is for a system command like a ZFS storage pool scrub.

```
# Minute Hour Day(M) Month Day(W) Command
0 4 * * 0 /usr/sbin/zpool scrub my-storage-pool
```

# `crontab` Examples ii

**Example 3: Check disk space every 15 minutes and log the output.** The >> appends the output to a log file, and 2>&1 ensures that errors are also logged.

```
# Minute Hour Day(M) Month Day(W) Command
*/15 * * * * /usr/bin/df -h >> \
/home/student/logs/disk_space.log 2>&1
```

# Advanced Bash & Scripts

# Redirection: Saving Output with >

Don't want to see output on the screen? Save it to a file with `>`.

**Warning:** This **overwrites** the file if it already exists.

**Example:** Save a list of your home directory contents to a file.

```
$ ls -l ~ > my_files.txt
```

# Redirection: Appending Output with >>

To **add** output to the end of a file without deleting its contents, use >>.

- This is great for creating log files.

**Example:** Add a timestamped entry to a log file.

```
$ echo "System rebooted at $(date)" >> system.log
```

# The Power of the Pipe |

The **pipe** is one of the most powerful concepts in the shell. It sends the output of one command to be the input of the next.

**Think of it as plumbing:** Command A -> | -> Command B

**Example:** Find all `.log` files in a directory.

```
# The output of 'ls' is "piped" to
# 'grep' to be filtered.
$ ls /var/log | grep .log
```

# I/O Streams: Stdin, Stdout, and Stderr

Every command in Linux uses three standard data streams.

- **`stdin` (0):** Standard Input. Usually your keyboard.
- **`stdout` (1):** Standard Output. Usually your screen.
  - Redirect with > (overwrite) or >> (append).
- **`stderr` (2):** Standard Error. Where error messages are sent.
  - You can redirect errors separately: `command 2> errors.log`.
- **Combining them:** To save both output and errors to one file, use `2>&1`.
  - `$ ./script.sh > all_output.log 2>&1`

# Your Environment: Variables

The shell uses variables to store information. By convention, they are in `ALL_CAPS`.

- `$HOME`: Your home directory.
- `$USER`: Your username.
- `$PATH`: A list of directories where the shell looks for programs.

**Example:** See the contents of the `$PATH` variable.

```
$ echo $PATH
/usr/local/sbin:/usr/local/bin:\
/usr/sbin:/usr/bin:/sbin:/bin
```

# Customizing Your Shell: `.bashrc`

The `~/.bashrc` file is a script that runs every time you open a new terminal. This is the place to personalize your shell.

You can edit it with a text editor:

```
$ nano ~/.bashrc
```

**Remember:** Changes won't apply until you open a new terminal or run `source ~/.bashrc`.

# Wildcards (Globbing)

Wildcards allow you to select groups of files based on patterns.

- *** (Asterisk):** Matches **any** number of characters (including zero).
  - `*.txt`: All files ending in `.txt`.
  - `data_*`: All files starting with `data_`.
- **? (Question Mark):** Matches exactly **one** character.
  - `file?.txt`: Matches `file1.txt`, `fileA.txt`, but *not* `file10.txt`.

**Example:** List all JPG images.

```
$ ls *.jpg
```

# Customization Example: Aliases

An **alias** is a shortcut or nickname for a longer command. They save you a lot of typing!

- Add this line to your `~/.bashrc` file:

```
alias ll='ls -alF'
```

- Now, when you type `ll` in a new terminal, bash will run `ls -alF` for you.

# Introduction to Bash Scripting

A script is simply a text file containing a sequence of commands.

1. The first line **must** be `#!/bin/bash`. This is called a "shebang."
2. Add your commands.
3. Use # for comments to explain your code.
4. Make the file executable with `chmod +x`.

# Scripting Example 1: Hello World

This script uses a variable and the echo command. It's the "Hello, World!" of scripting.

**File: `hello.sh`**

```bash
#!/bin/bash
# A simple hello world script

NAME="Student"
echo "Hello, $NAME!"
```

**To run it:**

```
$ chmod +x hello.sh
$ ./hello.sh
```

# Scripting Example 2: Using `if`

This script uses an `if` statement to check if a file exists before trying to use it.

**File: `check_file.sh`**

```bash
#!/bin/bash
# Checks for the existence of the system log file.

FILENAME="/var/log/syslog"

if [ -f "$FILENAME" ]; then
  echo "$FILENAME exists."
  # We could now do something with the file, e.g.
  # tail -n 5 "$FILENAME"
else
  echo "Warning: $FILENAME not found."
fi
```

# Scripting Example 3: Looping Over Files

A `for` loop lets you perform an action on a list of items, like files.

**File: `add_prefix.sh`**

```
#!/bin/bash
# Adds "backup_" prefix to all .txt files.
for file in *.txt
do
  # Check if it's a file before moving it
  if [ -f "$file" ]; then
    mv -- "$file" "backup_$file"
    echo "-> backup_$file"
  fi
done
echo "Batch rename complete."
```

# Scripting Example 5: Complex Script

```bash
#!/bin/bash
# Backs up specified items into a .tar.gz archive.
# Exit if no arguments are provided.
if [ "$#" -eq 0 ]; then
  echo "Usage: $0 <file1> <dir1> ..."
  exit 1
fi
DEST="$HOME/backups"
TIME=$(date +%Y-%m-%d_%H%M%S)
ARCHIVE="$DEST/$TIME-backup.tar.gz"
mkdir -p "$DEST" # Create backup dir if needed
echo "Creating archive..."
# "$@" holds all command-line arguments.
tar -czf "$ARCHIVE" "$@"
echo "Backup complete: $ARCHIVE"
```

## Scripting Example 6: Automating System Reports

You can combine system information commands into a single Bash script for easy reporting.

**File: sys_report.sh**

```
#!/bin/bash
# Automates the collection of system data into a re

REPORT_FILE="system_report_$(date +%Y%m%d).txt"

echo "--- LINUX SYSTEM REPORT ---" > "$REPORT_FILE"
echo "Generated on: $(date)" >> "$REPORT_FILE"
echo "User: $(whoami)" >> "$REPORT_FILE"

echo -e "\n[CPU INFO]" >> "$REPORT_FILE"
grep "model name" /proc/cpuinfo | head -n 1 >> "$RE

echo -e "\n[MEMORY USAGE]" >> "$REPORT_FILE"
free -h >> "$REPORT_FILE"
```

# Theory to Practice

You've now seen the core concepts of the Linux command line:

- **Navigating** the filesystem.
- **Managing** files, permissions, and users.
- **Combining** commands with pipes and redirection.
- **Automating** tasks with shell scripts.

Now, let's apply this knowledge in the practical part of the class.

# Support & Further Resources

Bookmark these pages. They are incredibly useful references.

- **Linux Terminal Cheat Sheet:** https://www.geeksforgeeks.org/linux-unix/linux-commands-cheat-sheet/
- **Bash Cheat Sheet:** https://github.com/RehanSaeed/Bash-Cheat-Sheet
- **Bash Scripting Cheat Sheet:** https://developers.redhat.com/cheat-sheets/bash-shell-cheat-sheet